

REFRIGER CHARGING MACHINE

Operation Scenario

Design Reference Document

PLC: Mitsubishi Q03UDV (Q mode) | GX Works2 IL | HMI: LS IXP2-1200

Written: 2026-06-05 | Updated: 2026-06-06 | Re-planned Session 1

1. Equipment

Line	Gun	Function	Steps
L1	A	Refrig injection	M10-M18 (9)
L2	B	Refrig injection	M30-M38 (9)
L3	-	Oil injection (per-gun)	M50-M53 (4)

- Only one line operates at a time (serial operation).
- L3 is per-gun oil (M65/M66 Gun A, M75/M76 Gun B). H+L solenoid structure only.

Solenoid Assignment

Signal	GUN A (L1)	GUN B (L2)
VAC SOL	M60 (Y020)	M70 (Y023)
STEM SOL	M61 (Y021)	M71 (Y024)
REFRIG FAST SOL	M62 (Y022)	M72 (Y025)
REFRIG BASE SOL	M63	M73
EXHAUST SOL	M64 (Y024)	M74 (Y027)
OIL FAST SOL	M65 (Y025)	M75 (Y028)
OIL BASE SOL	M66 (Y026)	M76 (Y029)
LINE VAC SOL	M68 (Y027, N/O)	
HYDRO PUMP	M96 (Y045, toggle)	
OIL LINE SOL	M97 (Y046)	
VACUUM PUMP	M67 toggle	
BUZZER / LAMPS G/R/Y	M69 / M77/M78/M79	

2. Power-On Default State

Setting	Value	Flag
Mode	MANUAL	M200=ON, M201=OFF
Active Gun	A	M210=ON, M211=OFF
Barcode	Used	M520=ON
Oil+Refrig	Enabled	M521=ON
Interlock	Used	M522=ON
Steps	IDLE	M10/M30/M50=ON
Hydro Pump	OFF	M96=OFF, M97=OFF

3. Step Machine

#	L1	L2	L3	Step Name
1	M10	M30	M50	IDLE
2	M11	M31	-	PRECHECK
3	M12	M32	-	GUN VAC
4	M13	M33	-	UNIT VAC
5	M14	M34	-	VAC CHECK
6	M15	M35	M51	REFRIG/OIL FAST
7	M16	M36	M52	REFRIG/OIL BASE
8	M17	M37	M53	EXHAUST/OIL COMPLETE
9	M18	M38	-	COMPLETE

- Offset: L1->L2=+20, L1->L3=+40. Warmup T0 0.5s before every step.
- Oil step M51/M52 auto-activates M96(HYDRO PUMP)+M97(OIL LINE SOL).
- Oil step BLOCKED when M304(HYDRO PUMP TRIP)=ON.

Step OK/Fail Flags

L1	L2	L3	Meaning
M100	M116		PRECHECK OK
M101	M117		GUN VAC OK
M102	M118		UNIT VAC OK
M103	M119		VAC CHECK OK
M105	M121		REFRIG BASE OK
M106	M122		EXHAUST OK

M107	M123		COMPLETE reached
		M145	OIL FAST OK
		M146	OIL BASE OK
		M147	OIL COMPLETE
M108	M124		GUN VAC NG
M109	M125		UNIT VAC NG
M111	M127		VAC CHECK NG

4. Auto Mode Sequence

```

IDLE(M10) -> START(M408)+Gun+T0 -> PRECHECK(M11):
  M80 interlock AND(D0>0) AND(D12>=1) -> M100
GUN VAC(M12): T1>=D2 -> M101 | NG:M108 | T0
UNIT VAC(M13): T2>=D4 AND D30<=D22 -> M102 | NG:M109 | T0
VAC CHECK(M14): VAC+STEM+LINE VAC ON, delta<=D24 -> M103 | NG:M111 | T0
IF M521=ON AND D18>=1 AND NOT M304: OIL chain
  OIL FAST(M51): HSC>=D16-D17 -> M145, 30s T13, activates M96/M97
  OIL BASE(M52): HSC in[D18-D19+-D20] -> M146/M351 NG, 30s T14
  OIL COMPLETE(M53) -> T0, RST M96/M97
IF M521=OFF OR D18<1 OR M304: skip oil
REFRIG FAST(M15): HSC>=D10-D11
REFRIG BASE(M16): HSC in[D12-D13+-D14] -> M105/M314 NG, 30s T4
EXHAUST(M17): T6>=D8 -> M106 | T0
COMPLETE(M18): SPC+cycle++, D1000=K1 -> IDLE, D0=0
  
```

L2 identical: M30-M38, M70-M74, D32-D63, M116-M127 flags.

5. Manual Mode Sequence

- Button press -> READY flag SET; press again -> RST (toggle cancel).
- START+READY -> T0 warmup via M453 -> single step only.
- OIL INJ blocked when M304(HYDRO PUMP TRIP) is ON.

Button	READY	Executes
GUN VAC (M403)	M220	M12/M32
UNIT VAC (M404)	M221	M13/M33
VAC CHECK (M405)	M222	M14/M34
REFRIG INJ (M406)	M223	M15->16 / M35->36
OIL INJ (M407)	M224	M51->52->53

6. Mode Control

```

SM402->SET M200,RST M201 (MANUAL)
M402 PLS M600 | M600&M200&M10->SET M201,RST M200
M600&M201&M10->SET M200,RST M201
  
```

7. Gun Select

```

SM402->SET M210,RST M211 (GUN A)
M400 PLS M601 | M601&M10&M30->SET M210,RST M211
M401 PLS M602 | M602&M10&M30->SET M211,RST M210
  
```

- Requires M10 AND M30 both ON (both lines IDLE).

8. STOP / EMG

Action	Behavior
STOP (M409)	M450/M451 latch -> RST steps -> EXHAUST -> IDLE. D1000=K6
NG Alarm	M108-111/M310-312->SET M450; M124-127/M330-332->SET M451
EMG M770 N/C	M452 self-hold -> RST all M10-M53+M60-M76+M96+M97 -> M410 reset

8-1. Hydro Pump

Item	Detail
Control	M96 (HYDRO PUMP) -> Y045
Oil Line Sol	M97 -> Y046, follows M96
HMI Button	M419 HYDRO PUMP ON/OFF (toggle)
Trip Input	M772 (X0A4, N/C)
Trip Alarm	M304 (self-holding, M410 reset)

Operation Logic

```
// HMI Toggle: M419 PLS M623
M623 & NOT M96 -> SET M96, SET M97
M623 & M96 -> RST M96, RST M97
```

```
// Auto ON during oil steps (pump NOT tripped)
M51 & NOT M304 -> SET M96, SET M97
M52 & NOT M304 -> SET M96, SET M97
```

- Two activation paths: HMI toggle OR automatic during oil steps.
- Oil charging step (M51/M52) is BLOCKED when M304=ON (hydro pump tripped).
- M304 stops both auto oil entry (ANI M304 in warmup chain) and manual oil entry.
- M96/M97 RST on: oil complete(M53), timeout, STOP, NG alarm, EMG(M452).
- M304 alarm: LDI M772 -> OR M304 ANI M410 -> OUT M304. Added to buzzer+RED lamp chain.

9. HMI Buttons

All buttons momentary, edge-detect via PLS M600-M623.

Button	Device	Action
GUN A	M400	Gun select (M10&M30 IDLE)
GUN B	M401	Gun select (M10&M30 IDLE)
AUTO/MANUAL	M402	Mode toggle (M10 IDLE)
GUN VACUUM	M403	Manual GUN VAC
UNIT VACUUM	M404	Manual UNIT VAC
VACUUM CHECK	M405	Manual VAC CHECK
REFRIG INJ	M406	Manual REFRIG INJ
OIL INJ	M407	Manual OIL INJ
START	M408	Initiate operation
STOP	M409	Halt -> EXHAUST -> IDLE
ALARM RESET	M410	Clear alarms + unmute
BUZZER STOP	M411	Mute buzzer (M500)
VAC PUMP ON/OFF	M412	Toggle M67
INTERLOCK EN	M413	Toggle M522
L1 COUNT RESET	M414	D240=0
OIL+REFRIG EN	M415	Toggle M521
L2 COUNT RESET	M416	D242=0
L1 USAGE RESET	M417	D200-D201=0
L2 USAGE RESET	M418	D220-D221=0
HYDRO PUMP	M419	Toggle M96/M97

10. HMI Lamp Backlight

Lamp	ON Condition
GUN A/B	M210/M211 selected
AUTO/MANUAL	M201/M200 active
GUN/UNIT/VAC/REFRIG/OIL	READY OR step active
GREEN (M77)	Running + no alarm
RED (M78)	Any alarm latch active (incl M304)
YELLOW (M79)	IDLE + interlock not OK

11. Alarm System

Shared Alarms

Alarm	Bit	Trigger
EMG Stop	M300	M770 N/C OPEN
Door Open	M301	M771 OFF (gated by M522)
Bombe Low	M302	Usage >= Bombe setting
Pump Trip	M303	Vacuum pump fault
Hydro Pump Trip	M304	M772 N/C OPEN -- BLOCKS OIL STEPS

L1 / L2 / L3 Alarms

- L1: M310(GUN TO), M311(UNIT TO), M312(VAC LEAK), M313(INJ TO 30s), M314(AMOUNT NG), M316/M317(PRESS), M318(TEMP), M319/M320(MODEL/TARGET)
- L2: M330-M340 (same offset +20)
- L3: M350(OIL TO 30s), M351(OIL AMOUNT NG), M304(HYDRO PUMP TRIP)

Buzzer

```
any_alarm AND NOT M500 -> OUT M69
M411 -> SET M500(mute)
new_alarm_edge -> RST M500
M410 -> RST M500(all clear)
```

12. Interlock

- L1: M81-M85 AND->M80. L2: M91-M95 AND->M90. M85/M95=SM400(spare).
- Violation during vac steps->RST step->EXHAUST. M301 gated by M522.

13. Injection Quantity

- 25 models/gun, 9 words/row. Gun A: D300-D524, Gun B: D550-D774.
- Barcode: PC writes D6980-D6999->table scan->D0/D32. Real target=Base+User Correction.

14-16. Flow Meter / Solenoids / DI

- HSC counts during M15-M16(refrig) and M51-M52(oil). 30s timeout (K3000).
- DI: EMG X0A2, Door X0A3, Interlock L1 X0AC-X0AF, Interlock L2 X0B7-X0BB.
- Output: Y020-Y046 covering all solenoids, hydro pump(M96->Y045), oil line sol(M97->Y046).

17. Analog Inputs

L1	L2	Signal	Width	EU L1	EU L2
V	V	Pressure	16-bit	D26	D58
V	V	Temperature	16-bit	D28	D60
V	V	Vacuum	32-bit	D30-D31	D62-D63

18. SPC & Counters

Item	L1	L2	Note
Cumulative usage	D200-D201	D220-D221	32-bit
Cycle count	D202	D222	
Cumulative inj	D204-D205	D224-D225	32-bit
Injection count	D240	D242	Resettable
Bombe alarm	D210-D211	D230-D231	32-bit
Stopwatch	D244 (0.1s)		Auto cycle only

19. D-Register Map

Parameter	L1	L2	Unit
Model#	D0	D32	-
Gun vac time	D2	D34	sec
Unit vac time	D4	D36	sec
Vac check time	D6	D38	sec
Exhaust time	D8	D40	sec
Refrig fast stop	D10-D11	D42-D43	g
Refrig target	D12-D13	D44-D45	g
Refrig tolerance	D14	D46	g
Oil fast stop	D16-D17	D48-D49	g
Oil target	D18-D19	D50-D51	g
Oil tolerance	D20	D52	g
Unit vac setting	D22	D54	Torr
Vac check setting	D24	D56	delta
Pressure	D26	D58	-
Temperature	D28	D60	-
Vacuum	D30-D31	D62-D63	Torr
Result Code	D1000	D1200	K1=OK,K6=STOP

20. File Structure

Module	Generated by	Content
MAIN.csv	gen_main_final.py	Step machine, mode/gun, interlock, STOP/EMG, lamps, hydro pump(M419, M623)
alarm.csv	gen_alarm.py	Alarms(shared+L1+L2+L3+M304), buzzer M69, mute M500
oilinj.csv	gen_oilinj.py	Oil injection: per-gun, HSC, M96/M97 auto on M51/M52, RST on complete/timeout
refinj/gunvac/unitvac/vac	gen_*.py	Refrig injection, vacuum steps
gmes.csv	gen_gmes.py	PC communication + SPC logging
idata.csv	gen_idata.py	DI->M, M->DO(Y020-Y046), HMI buttons M400-M419
indexs.csv	gen_indexs.py	Barcode copy + model lookup (25-model scan)

21. Key Design Rules

Rule	Detail
State bits	Self-holding: LD set_cond OR self ANI release_cond OUT self
Exclusive pairs	SET/RST: mode(M200/M201), gun(M210/M211), mute(M500), ready(M220-M224), enable flags, hydro pump(M96)
HMI buttons	Momentary 1-scan, edge-detect via PLS M600-M623
X/Y isolation	idata.py only. All logic references M/D registers.
Solenoid	H+L only (FAST+BASE). Single-sol: leave BASE unwired.
Warmup	T0 0.5s before EVERY step (auto: M460-M469, manual: M453)
STOP	All modes. M450/M451 latch->EXHAUST->IDLE. Release on IDLE.
EMG	M452->RST all steps+solenoids including M96/M97. Manual M410 reset.
Hydro Pump	M96/M97: HMI toggle(M419) OR auto ON during M51/M52. M304 blocks oil steps. RST on complete/timeout/STOP/EMG.
Lamps	M77(RUN=steps+!alarm), M78(RED=any alarm incl M304), M79(YELLOW=IDLE+!interlock)